
PAPER001

PREPRINT

Eoin Murray
Department of Engineering
University of Cambridge

Timothy O’Leary
Department of Engineering
University of Cambridge

June 29, 2026

ABSTRACT

1 Introduction

2 Results

2.1 The model: COBA baseline and the PING loop

2.2 Gamma onset across the $W_{ei} \times W_{ie}$ plane

2.3 Trained PING matches COBA accuracy at $\approx 10\times$ fewer spikes

2.4 Sparsity is supplied by the architecture, not learned

2.5 The gamma clock sets the excitatory firing rate

2.6 Dynamics and robustness of the gating

2.7 Streaming classification on continuous input

3 Discussion

4 Methods

4.1 Single-neuron and synapse dynamics

4.2 Network architecture

4.3 Mean-field reduction

4.4 Training

4.5 Measurement and analysis

4.6 Integration and parameters

4.7 Datasets and evaluation

4.8 Reproducibility

4.9 Software and implementation

4.10 Generative-AI use

Code and data availability

A Figures

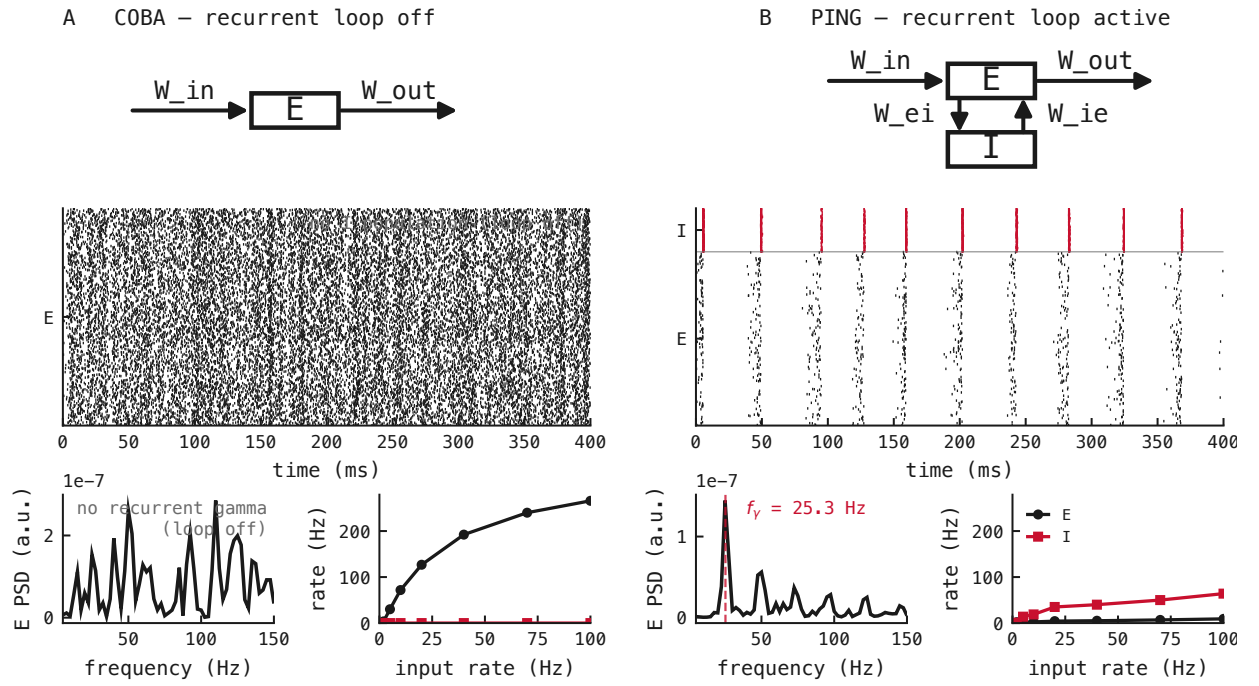


Figure 1:

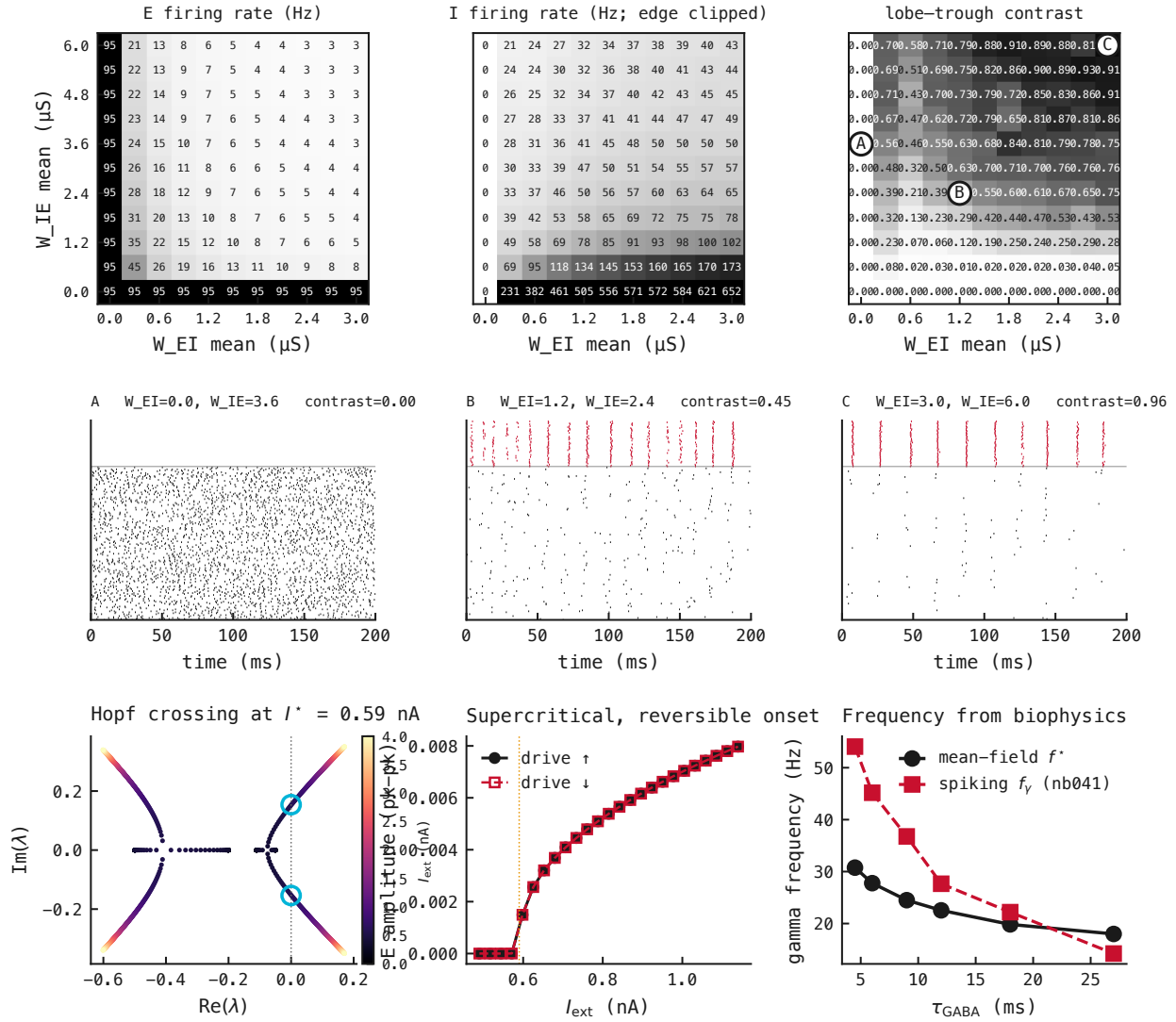


Figure 2:

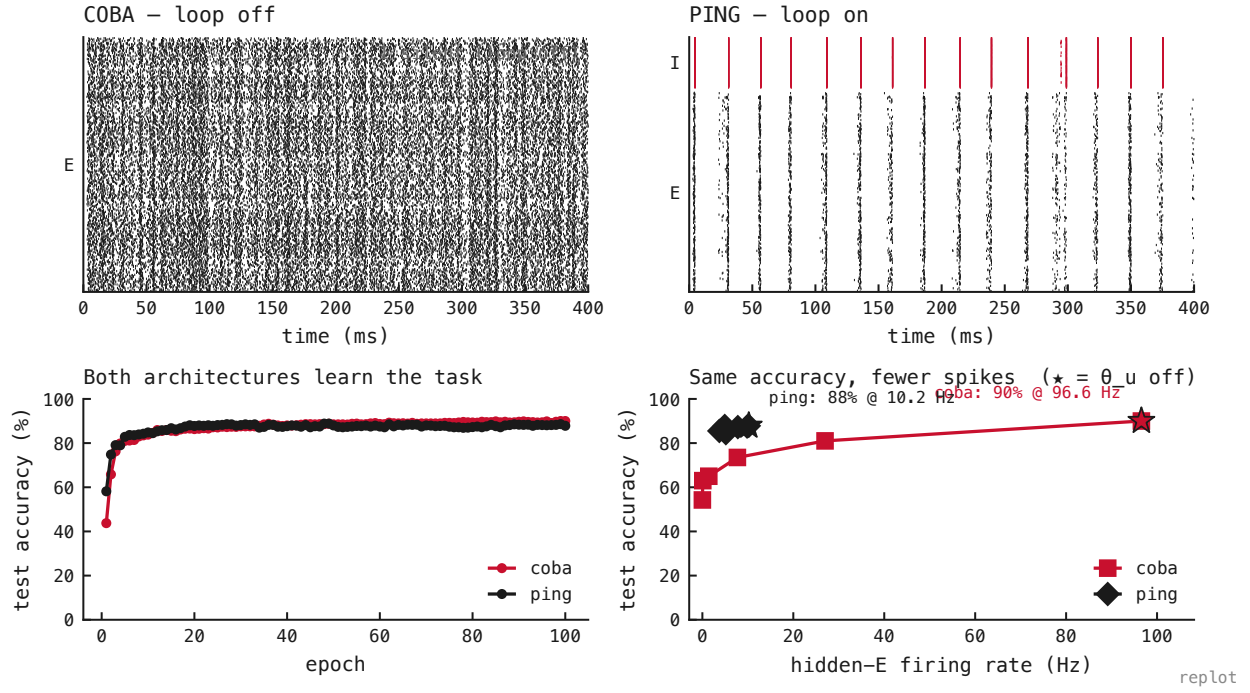


Figure 3:

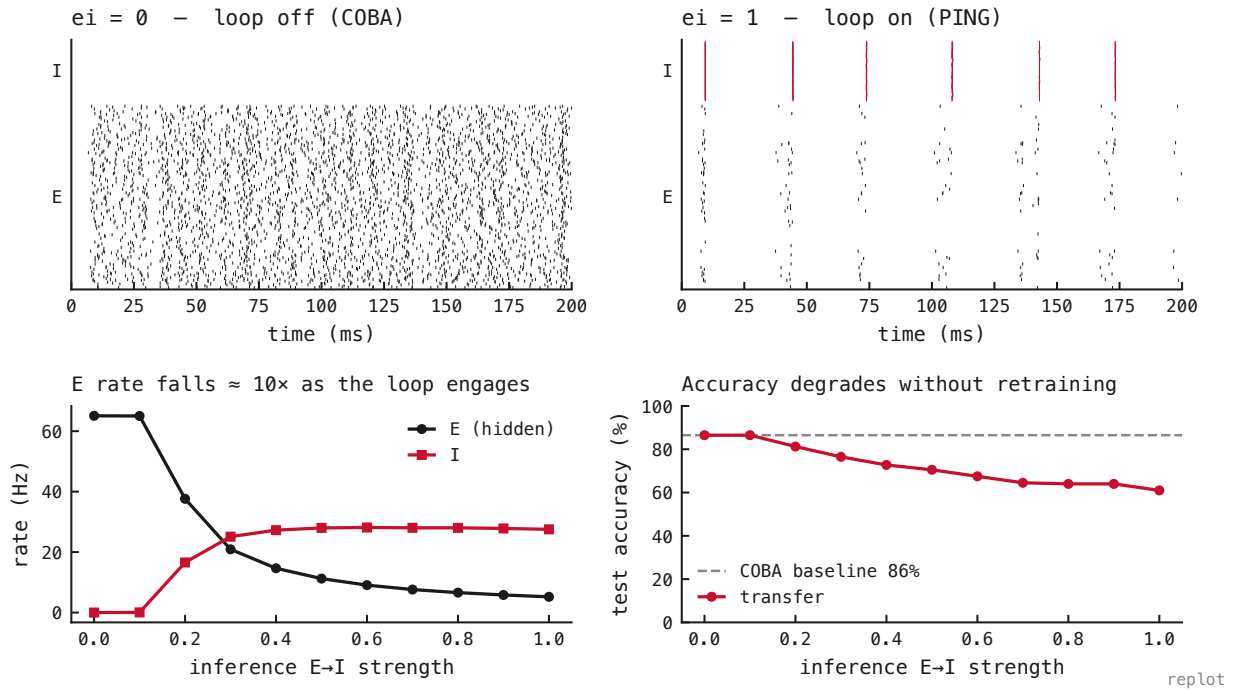


Figure 4:

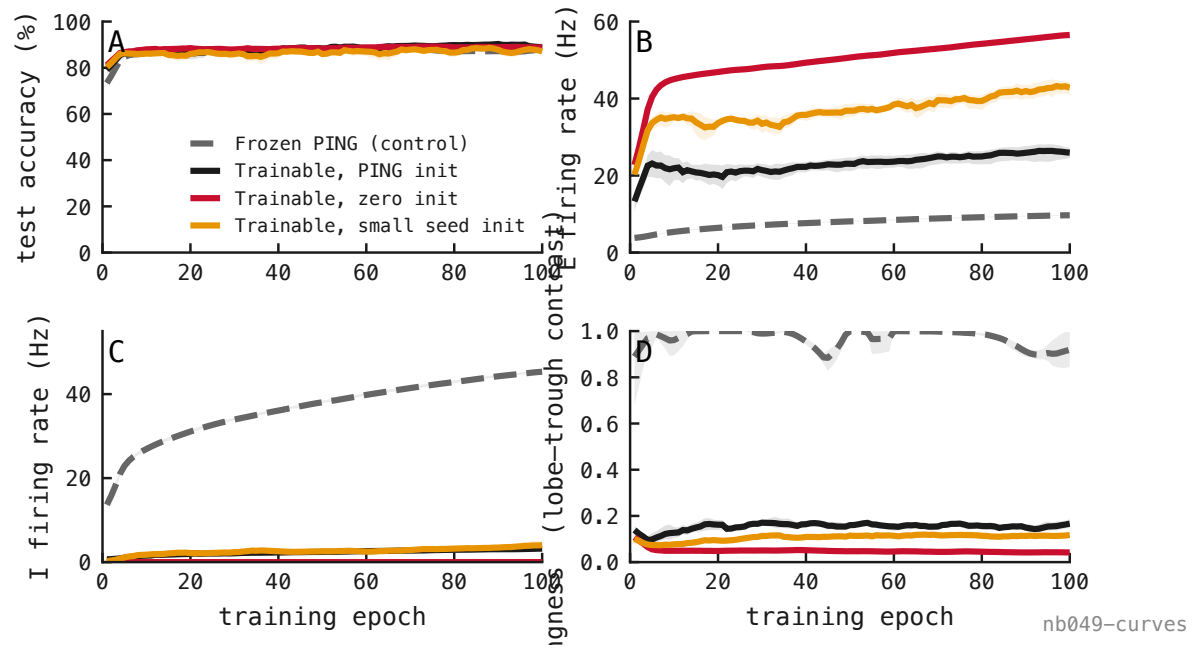
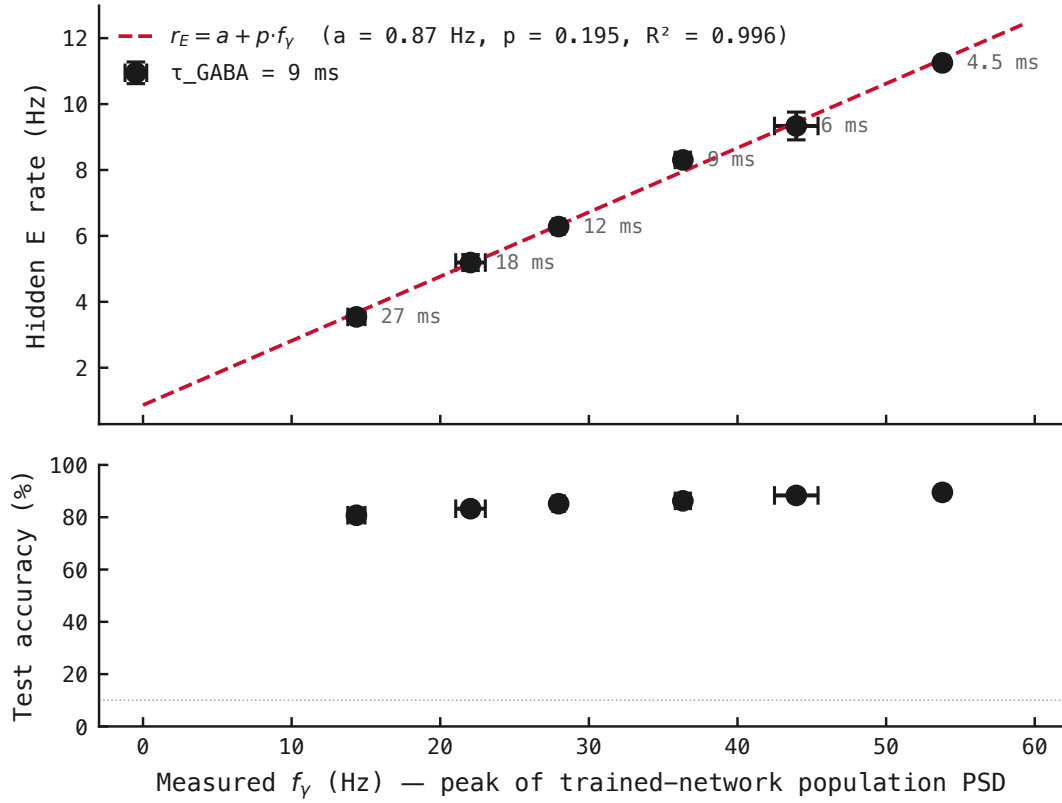


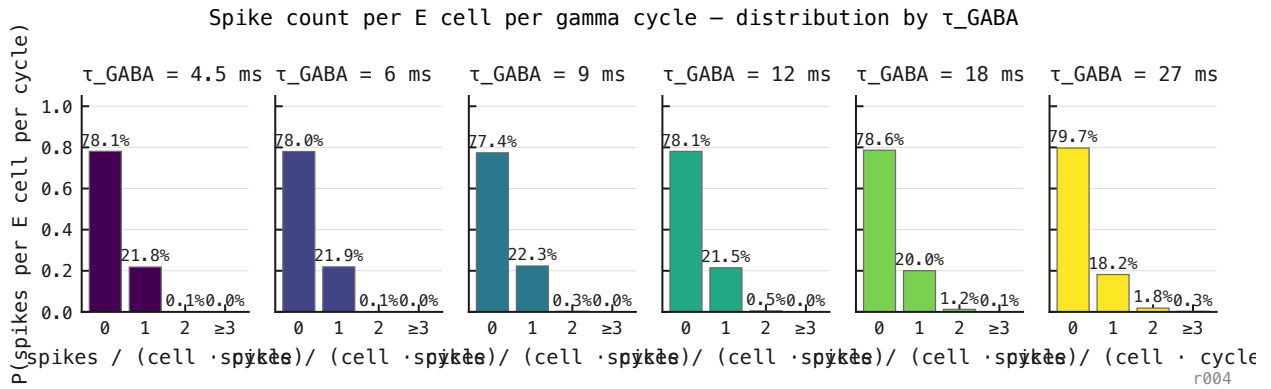
Figure 5:

Retrained τ_{GABA} sweep — post-training E rate tracks measured f_{γ}



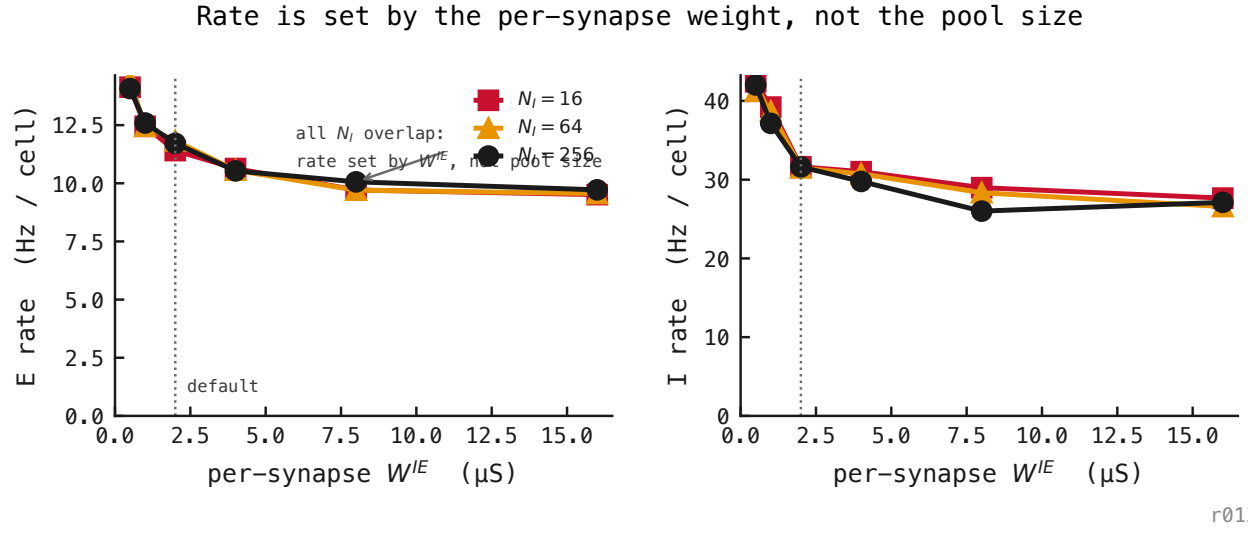
r010

Figure 6:



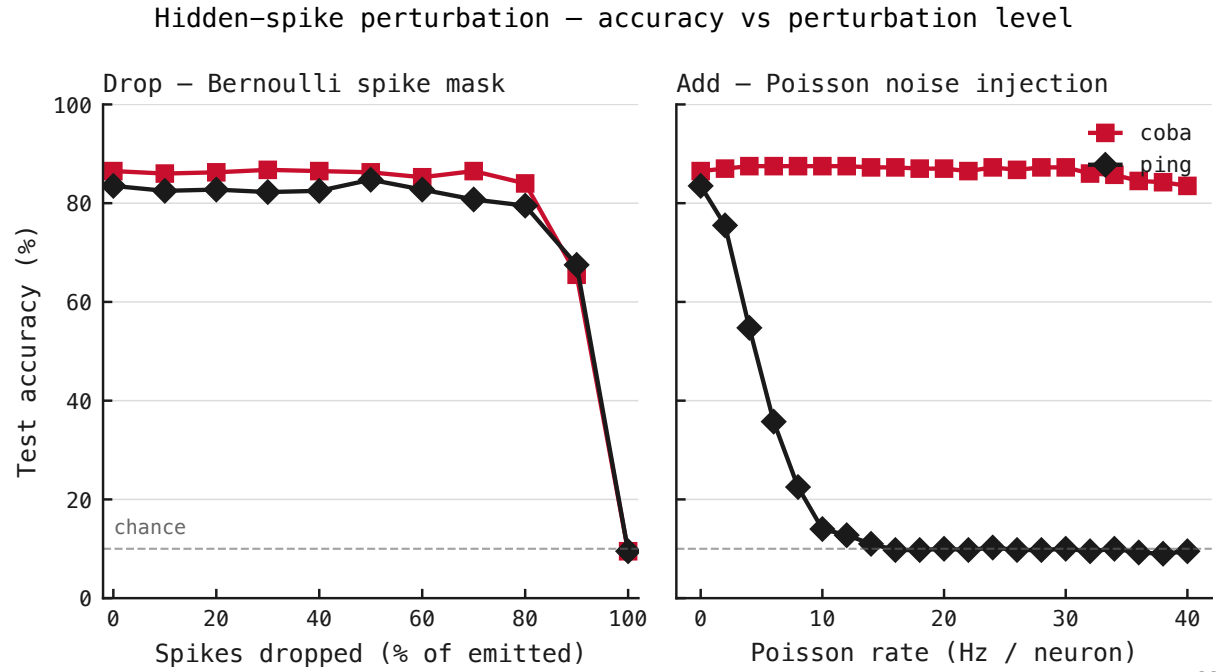
r004

Figure 7:



r013

Figure 8:



r002

Figure 9:

Gamma gates the rate, not mean inhibition – matched mean I, opposite E response

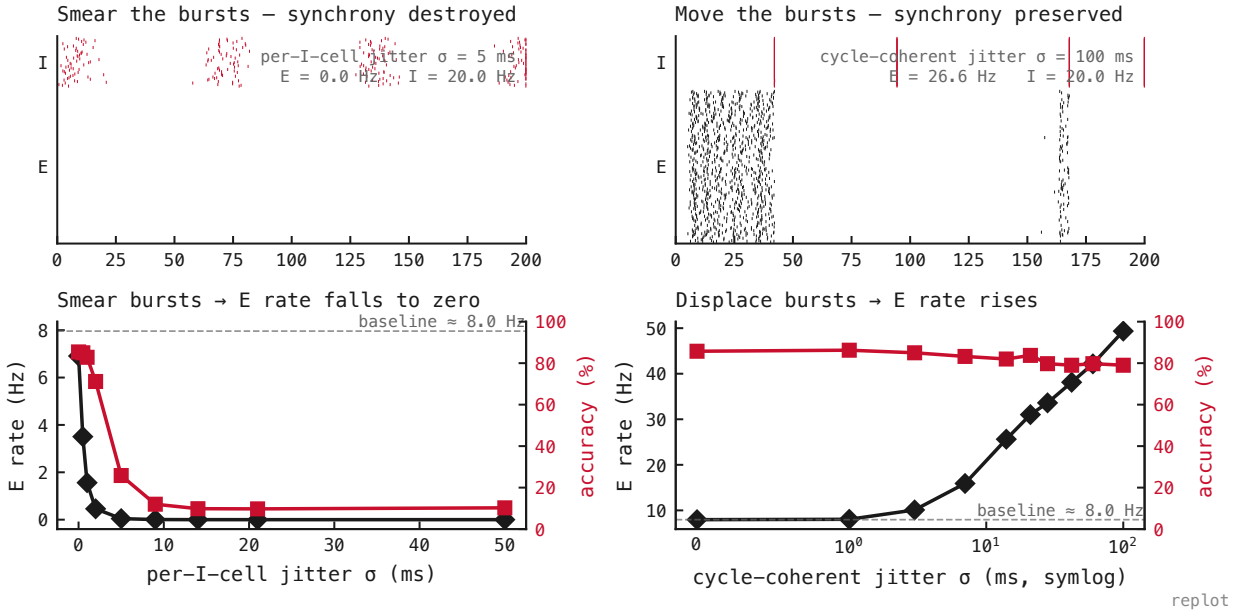


Figure 10:

PING Δt audit – post-training E rate and accuracy vs integration timestep

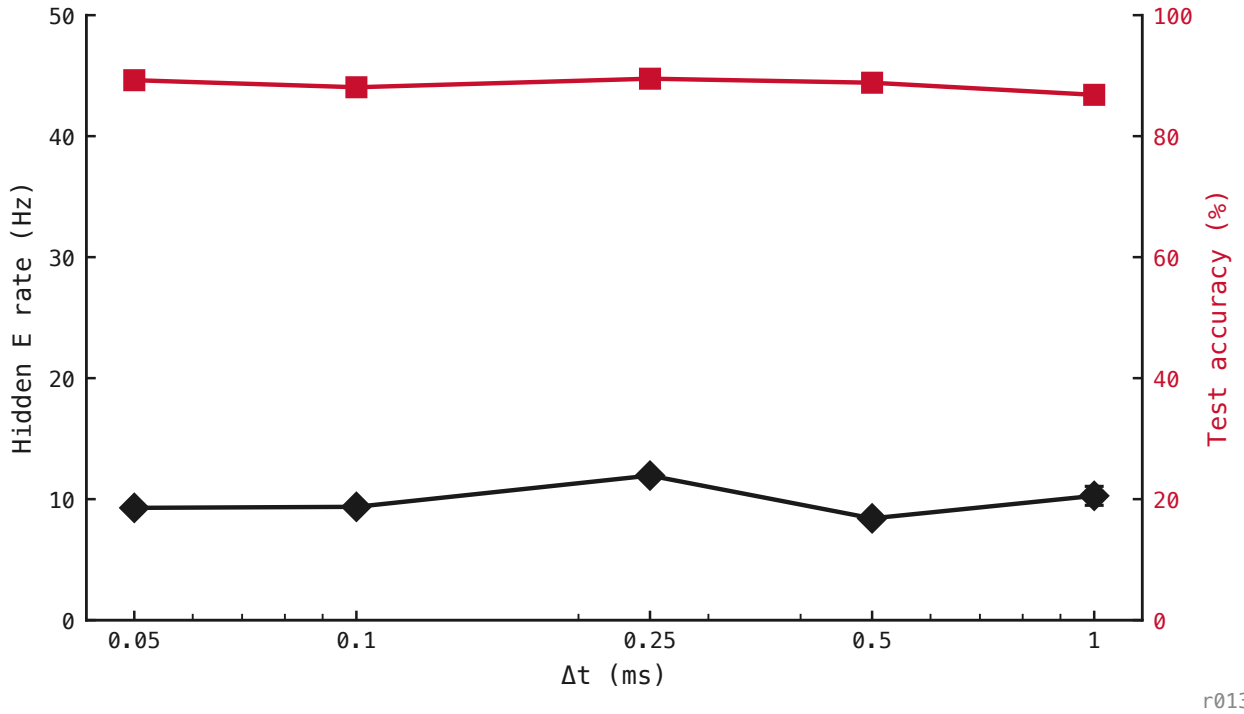
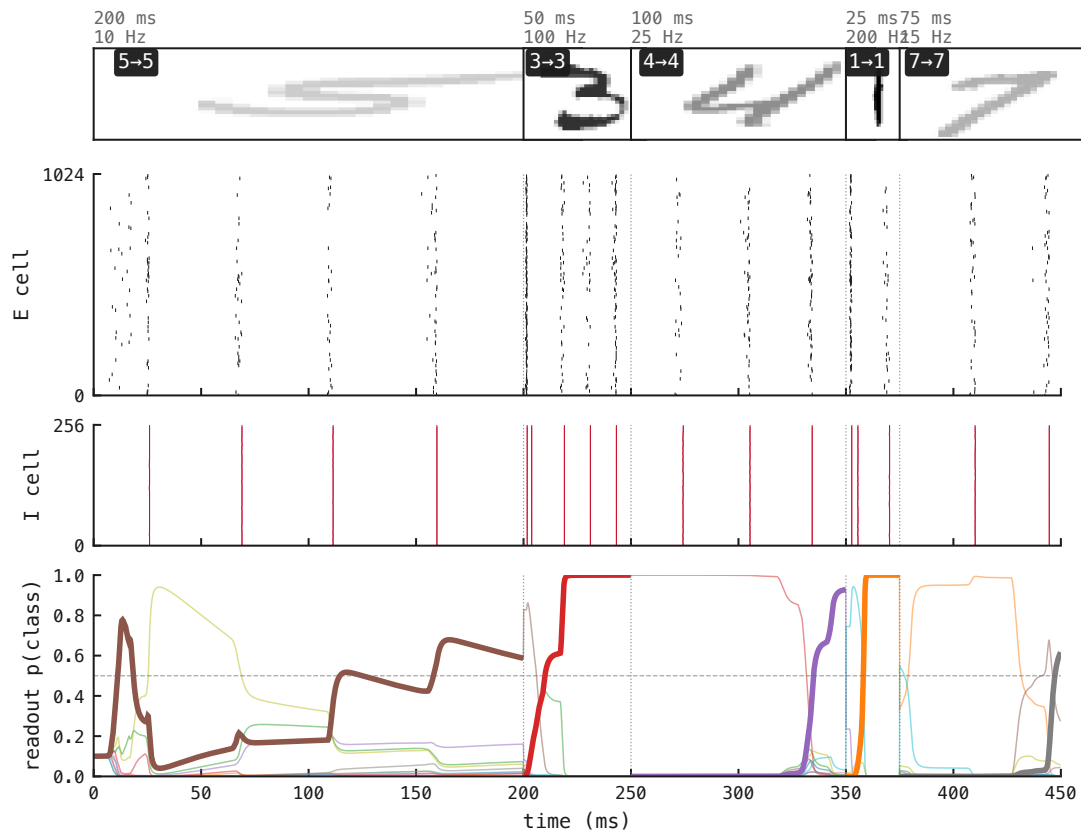


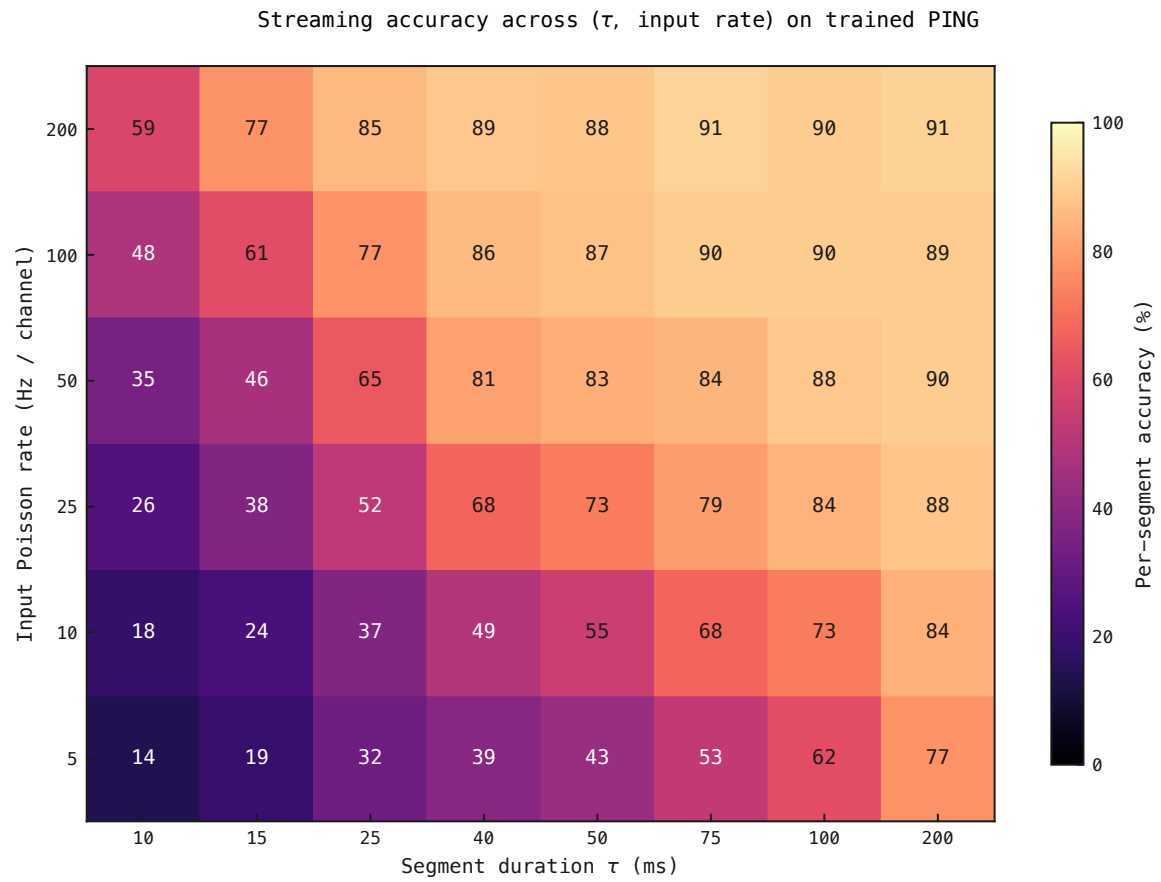
Figure 11:

Streaming digit classification — per-segment (τ , rate) varies, trained PING (no retraining)



r004

Figure 12:



r004

Figure 13:

References